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ChE 311 Midterm Exam II, Spring 2019

Closed book and closed notes. Only provided tables are allowed.

50 minutes, 100 points

1. (5 points) Which of the following statements is/are true?

(A) Eddy viscosity of Boussinesq is a property of the fluid.

(B) Eddy viscosity for a flow system is constant everywhere.

(C) Eddy viscosity of Boussinesq gives a first-order approximation for Reynolds Stress.

(D) None of above.

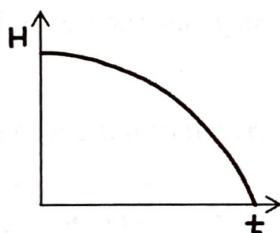
Answer : ~~A~~, C

2. (5 points) Which of the following curve represents the draining curve of a conical tank?

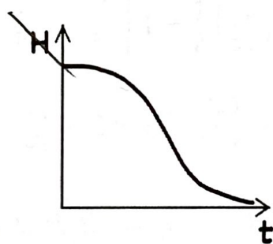


Answer : C

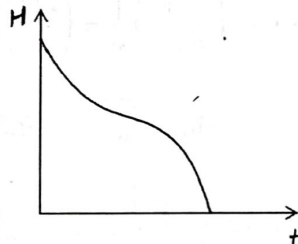
(A)



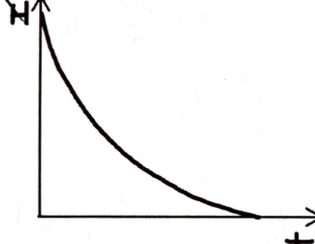
(B)



(C)



(D)



3. (5 points) Estimate the order of the Reynolds number of a swimmer who swims a kilometer in 20 minutes. Choose the closest answer.

(A) 0.01

(B) 1

(C) 100

(D) 100,000

Answer : D

$$Re = \frac{\rho V L}{\mu} = \frac{1025 \text{ kg/m}^3 \times \frac{1000 \text{ m}}{20 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} \times 5 \text{ m}}{8.9 \times 10^{-4} \text{ Pa}\cdot\text{s}}$$

$$= 4166.67 \text{ kg/m}\cdot\text{s}$$

4. (5 points) For flow in a circular tube, the coordinate was set up as z was the flow direction and r was the radial direction. Please write down the $\bar{\tau}_{rz}^{(t)}$ by its definition in terms of velocity fluctuation?

$$\bar{\tau}_{rz}^{(t)} = \rho \overline{v_z' v_r'}$$

5. (5 points) For above flow in a circular tube, please write down $\bar{\tau}_{rz}^{(t)}$ in terms of wall shear stress (τ_0) and the radius of the tube (R).

$$\bar{\tau}_{rz} = \tau_0 \left(1 - \frac{y}{R}\right) = \bar{\tau}_{rz}^{(v)} + \bar{\tau}_{rz}^{(t)}$$

$$\Rightarrow \bar{\tau}_{rz}^{(t)} = \tau_0 \left(1 - \frac{y}{R}\right) + \mu \frac{dv_z}{dr}$$

6. (30 points) **Turbulent flow in a circular tube.**

Please determine the ratio of $\mu^{(t)}/\mu$ at $y = R/3$ for water flowing at a steady rate in a long, smooth, circular tube under the following conditions.

Tube radius, $R = 10$ cm

Wall shear stress, $\tau_0 = 0.2$ Pa

Density of the fluid, $\rho = 1000$ kg/m³

Viscosity of the fluid, $\mu = 1 \text{ cp} = 10^{-3} \text{ Pa}\cdot\text{s}$

$$V^* = \sqrt{\frac{\tau_0}{\rho}} \quad v^+ = \frac{v_z}{V^*}$$

$$y^+ = \frac{y V^*}{\nu} \Rightarrow y = \frac{y^+ \nu}{V^*}$$

For viscous sublayer, $\frac{\bar{v}_x}{v_*} = \frac{y v_*}{\nu} \left[1 - \frac{1}{2} \left(\frac{\nu}{v_* B} \right) \left(\frac{y v_*}{\nu} \right) - \frac{1}{4} \left(\frac{y v_*}{14.5 \nu} \right)^3 + \dots \right] \quad 0 < \frac{y v_*}{\nu} < 5$

For inertial sublayer, $\frac{\bar{v}_x}{v_*} = 2.5 \ln \left(\frac{y v_*}{\nu} \right) + 5.5 \quad \frac{y v_*}{\nu} > 30$

$$\bar{\tau}_{rz}^{(t)} = \bar{\tau}_{rz}^{(v)} + \bar{\tau}_{rz}^{(t)} = -\mu \frac{dv_z}{dr} - \mu^{(t)} \frac{dv_z}{dr} = -\frac{dv_z}{dr} (\mu + \mu^{(t)})$$

$$\mu^{(t)} = -\frac{\bar{\tau}_{rz}}{\frac{dv_z}{dr}} - \mu$$

$$\frac{\mu^{(t)}}{\mu} = -\frac{1}{\mu} \frac{\tau_0 \left(1 - \frac{y}{R}\right)}{\frac{dv_z}{dy}} - 1 = \frac{1}{\mu} \frac{\tau_0 \left(1 - \frac{y}{R}\right)}{\frac{dv_z}{dy}} - 1$$

$$= \frac{\rho V^{*2} \left(1 - \frac{y}{R}\right)}{\mu \frac{dv_z}{dy}} - 1 = \frac{V^{*2} \left(1 - \frac{y}{R}\right)}{\nu \frac{dv_z}{dy}} = \frac{V^{*2} \left(1 - \frac{y}{R}\right)}{\frac{\nu^2}{V^*} \frac{dv^+}{dy^+}} = \frac{1 - \frac{y}{R}}{\frac{dy^+}{V^*}} \quad \checkmark$$

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$$y = \frac{2V^*}{J} = \frac{\frac{R}{3} \cdot \sqrt{\frac{\tau_0}{\mu}}}{\frac{\mu}{\rho}} = \frac{\frac{0.1 \text{ m}}{3} \cdot \sqrt{\frac{0.2 \text{ Pa} \cdot \frac{\text{kg/m}^3}{12}}{1000 \text{ kg/m}^3}}}{\frac{10^{-3} \text{ Pa} \cdot \text{s}}{1000 \text{ kg/m}^3}} = \frac{\frac{0.1 \text{ m}}{3} \cdot 0.0141 \text{ m}}{1 \times 10^{-5} \text{ m}^2 \cdot \text{s}^{-1}}$$

$$= 471.408$$

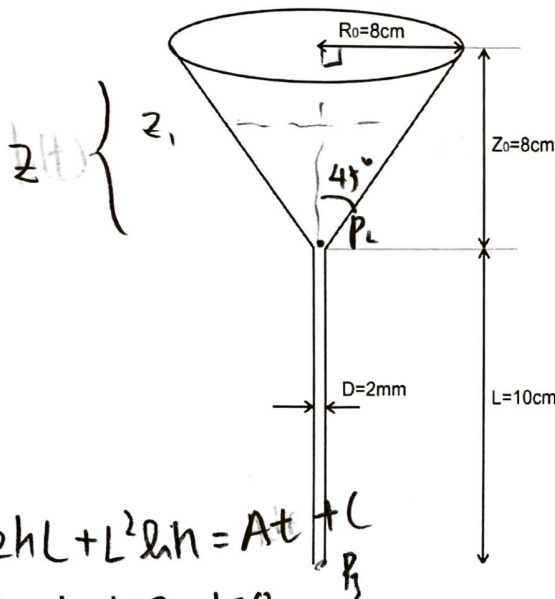
Since y is greater than z_0 , $\frac{dv^*}{dy^*} = \frac{2.5}{y^*} = 0.00530$

$$\frac{\mu(t)}{\mu} = \frac{1 - \frac{z}{R}}{\frac{dv^*}{dy^*}} = \frac{1 - \frac{1}{3}}{0.00530} = \boxed{125.708} \quad \checkmark$$

$$\frac{\text{kg m}^{-1} \cdot \text{s}^{-1}}{\text{kg/m}^3} = \text{m}^2 \text{s}^{-1}$$

7. (25 points) Using a conical funnel to measure the viscosity of a fluid.

The dimensions of the conical funnel are shown in the figure. The initial liquid level is at $Z_1 = 5 \text{ cm}$. The efflux time for completely draining the fluid (t_{efflux}) is 100s. $R_0 = Z_0 = 8 \text{ cm}$. The pipe diameter (D) is 2mm. Pipe length (L) is 10 cm. Density of the fluid (ρ) has been measure, which is 1500 kg/m^3 . Take $g = 10 \text{ m/s}^2$. Please decide the viscosity of the fluid.



$$\frac{dm}{dt} = -w_2 = -Q_2 \rho = -\int \frac{\pi \Delta p R_p^4}{8 \mu L}$$

$$V = \int_0^Z \pi r^2 dz = \int_0^Z \pi \cdot \frac{z^2}{Z_0^2} R_0^2 dz$$

$$\left(\frac{r}{R_0} = \frac{z}{Z_0} \Rightarrow r = \frac{z}{Z_0} R_0 \right)$$

$$= \frac{\pi R_0^2}{Z_0^2} \frac{z^3}{3}$$

$$\frac{dm}{dt} = \frac{d}{dt} \rho V = -\int \frac{\pi \Delta p R_p^4}{8 \mu L}$$

$$= \frac{d}{dt} \left(\rho \frac{R_0^2}{Z_0^2} \frac{z^3}{3} \right) = -\int \frac{\pi \Delta p R_p^4}{8 \mu L}$$

$$\Rightarrow \frac{d}{dt} \left(\frac{R_0^2}{Z_0^2} \frac{z^3}{3} \right) = -\frac{\Delta p R_p^4}{8 \mu L}$$

$$\Delta p = P_2 - P_1 + \rho g L = \rho g (Z + L)$$

$$\Rightarrow \frac{d}{dt} \left(\frac{R_0^2}{Z_0^2} \frac{z^3}{3} \right) = -\frac{\rho g (Z + L) R_p^4}{8 \mu L}$$

$$\text{Set } h = Z + L \Rightarrow Z = h - L$$

$$\Rightarrow dh = dz$$

$$\Rightarrow \frac{z^2 dz}{Z + L} = -\frac{\rho g R_p^4}{8 \mu L} \cdot \frac{Z_0^2}{R_0^2} dt$$

$$\Rightarrow \frac{(h - L)^2}{h} dh = -\frac{\rho g R_p^4}{8 \mu L} \cdot \frac{Z_0^2}{R_0^2} dt$$

$$\Rightarrow \int \frac{h^2 - 2hL + L^2}{h} dh = \int A dt$$

$$\frac{h^2}{2} - 2hL + L^2 \ln h = At + C$$

$$\text{BC1: } @ h = L + Z_1, t = 0$$

$$\frac{(L + Z_1)^2}{2} - 2(L + Z_1)L + L^2 \ln(L + Z_1)$$

$$= C$$

$$= \frac{(0.15 \text{ m})^2}{2} - 2(0.15 \text{ m}) \cdot 0.1 \text{ m} + (0.1 \text{ m})^2 \ln(0.15 \text{ m})$$

$$= -0.037 \text{ m}^2$$

$$\text{BC2: } @ h = L, t = 100 \text{ s}$$

$$\frac{h^2}{2} - 2hL + L^2 \ln h = -\frac{\rho g R_p^4}{8 \mu L} \cdot \frac{Z_0^2}{R_0^2} t + C$$

$$-\frac{\rho g R_p^4}{8 \mu L} \cdot \frac{Z_0^2}{R_0^2} t = \frac{h^2}{2} - 2hL + L^2 \ln h - C$$

$$= \frac{(0.1 \text{ m})^2}{2} - 2 \cdot (0.1 \text{ m})^2 + (0.1 \text{ m})^2 \ln 0.1 - C$$

$$= -3.047 \times 10^{-4} \text{ m}^2$$

$$\frac{\rho g R_p^4}{8 \mu L} \cdot \frac{z_0^2}{R_0^2} t = 3.047 \times 10^{-4} \text{ m}^2$$

$$\Rightarrow \mu = \frac{\rho g R_p^4}{8 L} \cdot \frac{z_0^2}{R_0^2} t \cdot \frac{1}{3.047 \times 10^{-4} \text{ m}^2}$$

$$= \frac{1500 \text{ kg/m}^3 \cdot (0.001 \text{ m})^4 \times 10 \text{ m/s}^2}{8 \cdot 0.1 \text{ m}} \cdot 100 \text{ s} \cdot \frac{1}{3.047 \times 10^{-4} \text{ m}^2}$$

$$= 0.00615 \text{ kg m}^{-1} \text{ s}^{-1} = 0.00615 \text{ Pa} \cdot \text{s}$$